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Long-term safety and efficacy of amisulpride in subchronic or chronic schizophrenia

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Amisulpride is an atypical antipsychotic with selective affinity for dopamine D_2/D_3 receptors. In this long-term, open, randomised, multicentre trial, patients with chronic or subchronic schizophrenia received amisulpride (n = 370) or haloperidol (n = 118) for 12 months. Dosage regimens were flexible (amisulpride 200–800 mg/day, haloperidol 5–20 mg/day). Improvement in mean Brief Psychiatric Rating Scale total score was significantly greater for amisulpride than haloperidol (17.0 versus 12.8, P = 0.01). Positive symptoms (Positive and Negative Syndrome Scale [PANSS] positive) improved in a similar way in each group but amisulpride caused a significantly better improvement in negative symptoms (PANSS negative) (7.1 versus 3.7, P < 0.0001). Improvements in Global Assessment of Functioning (GAF) and Quality of Life Scale (QLS) scores were also significantly greater in the amisulpride group (GAF –20.1 versus –13.6, P = 0.001; QLS –0.64 versus –0.30, P = 0.02). Adverse events were mainly psychiatric in nature, and occurred with similar frequency in each group (amisulpride 254/370, 69%; haloperidol 82/118, 70%). Extrapyramidal symptoms were more frequent for haloperidol (48/118, 41% versus 96/370, 26% for amisulpride), leading to a greater requirement for antiparkinsonian medication (haloperidol 66/118, 56% versus amisulpride 118/370, 32%). Haloperidol significantly aggravated parkinsonism, akathisia and involuntary movement compared to amisulpride. The overall incidence of endocrine events was comparable between groups (4% for amisulpride, 3% for haloperidol). Maintenance of efficacy was comparable in both treatment groups; 59% of amisulpride patients and 55% of haloperidol patients improved after 1 month of therapy remained improved throughout the study period. Amisulpride is effective following flexible long-term administration and significantly improves social functioning and quality of life. Int Clin Psychopharmacol 15:13–22 © 2000 Lippincott Williams & Wilkins

Keywords: amisulpride, chronic schizophrenia, flexible administration, long-term safety, maintenance of efficacy

INTRODUCTION

The treatment of schizophrenia continues to represent a significant challenge because of the severe, chronic and disabling nature of the illness and the heterogeneity of symptoms (Kane, 1997). Onset gradually occurs in late adolescence or early adulthood and the lifetime prevalence of the disorder is estimated between 0.2% and 1% (DSM III-R; American Psychiatric Association, 1987). Most patients will require treatment for psychotic relapses or exacerbations throughout their lifetime. The positive symptoms of schizophrenia (Andreasen and Olsen, 1982), such as hallucinations, and delusions, can be managed effectively by antipsychotic medication in about 70% of cases (Kane, 1997) but the negative symptoms – apathy, attentional deficit, poverty of thought or speech, and social withdrawal – do not generally respond to the drugs currently available (Möller, 1993). Although antipsychotic agents are the most effective for treatment of acute episodes, their benefit is compromised by the adverse motor side-effects of long-term use and other neurovegetative side-effects inducing low compliance.

Amisulpride is a new atypical antipsychotic agent with selective affinity for dopamine D_2 and D_3 receptors (Sokoloff *et al.*, 1990; Perrault *et al.*, 1997). In animals, amisulpride has been shown to preferentially block presynaptic receptors and has a greater affinity for dopamine receptors in limbic structures than in the striatum (Perrault *et al.*, 1997; Schoemaker *et al.*, 1997). Furthermore, it does not provoke catalepsy in rats (Perrault *et al.*, 1997), a feature characteristic of striatal postsynaptic D_2 blockade. Clinical studies have shown that amisulpride is at least as effective as haloperidol (Möller *et al.*, 1997; Puech *et al.*, 1998) and risperidone (Fleurot *et al.*, 1997), and more effective than flupentixol (Wetzel *et al.*, 1998) in the short-term treatment of the positive symptoms of schizophrenia. Moreover, efficacy in patients with predominantly negative symptoms has also been demonstrated (Boyer *et al.*, 1995; Paillère-Martinot *et al.*, 1995; Loo *et al.*, 1997; Danion *et al.*, 1999), thus differentiating amisulpride from conventional and other newer neuroleptics.

Clinical experience to date indicates that amisulpride is well tolerated and has a favourable safety profile. In particular, amisulpride is associated with a lower incidence of extrapyramidal effects than haloperidol (Möller *et al.*, 1997; Puech *et al.*, 1998). In a previous long-term study in severely disabled hospitalised patients with chronic schizophrenia, amisulpride was as effective as haloperidol and was well-tolerated during administration for 1 year (Speller *et al.*, 1997). The study reported here was designed to assess the safety and maintained efficacy of amisulpride during long-term usage over a period of 12 months in schizophrenic patients with acute exacerbations followed-up as outpatients.

METHODS

Patients and study design

This was a randomised, open study involving two parallel groups, one receiving amisulpride, the other haloperidol, over a period of 12 months. Patients were randomised in a 3:1 ratio amisulpride to haloperidol; they were followed-up as outpatients but could be hospitalised in the initial phase of the study. The study included patients with subchronic or chronic schizophrenia with acute exacerbation (paranoid, disorganised or undifferentiated) meeting the criteria of the revised DSM III-R (American Psychiatric Association, 1987). Patients had to have a minimum score of moderate (4 on a seven-point scale) on at least two of the four Brief Psychiatric Rating Scale (BPRS; Overall and Gorham, 1962) positive items. As well as the standard exclusion criteria for clinical trials, patients with productive (positive) symptoms in whom the disorder did not correspond to a schizophrenic criteria: schizophreniform or schizoaffective disorder, maniac or excitative depressive states, and severe cardiovascular disease, Parkinson's disease or phaeochromocytoma were excluded from the study. The doses of amisulpride and haloperidol were adjusted according to efficacy and tolerance. During an acute episode, the maximum permitted daily dose of amisulpride was 1200 mg, and 30 mg for haloperidol. After stabilisation of the acute episode, the daily doses of amisulpride were to fall within the range 200–800 mg, and of haloperidol, 5–20 mg. Both treatments were given orally, twice daily. Other neuroleptic agents and psychotropic drugs such as antidepressants or thymoregulators were not permitted during the study, but diazepam (maximum dose 40 mg daily) and antiparkinsonian agents were authorised for the relief, respectively, of insomnia and/or anxiety and extrapyramidal symptoms. Patient compliance was assessed via regular pill-count by the investigator at each visit.

The study was conducted in accordance with the declaration of Helsinki (1964), revised in Tokyo (1975), Venice (1983) and Hong-Kong (1989). The protocol was approved by local ethics committee of the participating centre (or country, if applicable). All patients gave their written informed consent before start of the trial.

Assessment

The principal aim of the study was to assess the long term safety of amisulpride. Adverse events were reported throughout the study. At each visit (baseline = day 0 [D0], D7, D15, month 1 [M1], M3, M6, M9 and M12 [Dend]) blood pressure and heart rate were assessed and an electrocardiogram (ECG) was performed at baseline, M6 and at study end. Three scales were used to assess neurological safety. The 10-item Simpson–Angus extrapyramidal scale assessed parkinsonian symptoms (Simpson and Angus, 1970), while the 4-item Barnes Rating Scale (Barnes, 1989) was used to detect drug-induced akathisia and the 10-item Abnormal Involuntary Movement Scale (AIMS; National Institute of Mental Health, 1976a) for tardive dyskinesia. Patients were weighed at each visit, and laboratory parameters (blood biochemistry, haematology and prolactinaemia) were monitored throughout the study. All assessments were based on changes between baseline and the end of the study.

A secondary objective of the study was to determine the long term efficacy of amisulpride, using the BPRS, Global Clinical Impressions (CGI; National Institute of Mental Health, 1976b), Positive and Negative

Syndrome Scale (PANSS; Kay *et al.*, 1987), Global Assessment of Functioning (GAF; American Psychiatric Association, 1987) and Quality of Life Scale (QLS; Heinrichs *et al.*, 1984). The primary efficacy variable was the change in BPRS total score between baseline and study end.

Statistical analysis

All patients receiving at least one dose of study treatment after randomisation (the intent-to-treat population, ITT) were included in the analyses and the last-observation-carried-forward procedure was used to present results. Quantitative variables were analysed using one-way ANOVA and *t*-test. The chi-squared test, Cochran-Mantel-Haenszel test, Kaplan–Meier method and log-rank test were used to analyse qualitative variables. Analyses were performed using SAS software, version 6.08 (SAS Institute Inc., Cary, NC, USA)

RESULTS

Patient characteristics

Sixty-two centres in nine countries participated in the study. Between February 1993 and October 1996, 370 patients were randomised to receive amisulpride and 119 to haloperidol. One patient in the haloperidol group was excluded from further analysis, being lost to follow-up after randomisation and before commencement of treatment. The majority of the patients were Caucasian; their average age was 37.5 years (range 18–65 years) and two-thirds were male (Table 1). Paranoid schizophrenia was the most common type of schizophrenia encountered at baseline. The majority of patients were hospitalised with chronic disease (Table 1). Their mean BPRS score was 56.2. The majority of the patients had no or questionable symptoms of akathisia at baseline, and their mean Simpson–Angus and AIMS scores were low, at 0.37 and 2.5, respectively (Table 1). There was no significant difference between the treatment groups for the demographic and disease characteristics with the exception of age, duration of illness and AIMS total score which were significantly lower ($P = 0.015$, 0.026 and 0.049, respectively) in the amisulpride group.

In the amisulpride group, 167/370 (45%) patients were prematurely withdrawn from the study, as were 62/119 (52%) patients in the haloperidol group (Table 2). More than 60% of withdrawals occurred within the first 3 months of the study, and few patients were lost to follow-up (amisulpride, $n = 10$, 3%; haloperidol, $n = 6$; 5%).

Safety

During the first 3 months of the study, the mean daily dose of amisulpride was 626 ± 232 mg and of

Table 1. Baseline demographic and disease characteristics

Characteristics	Amisulpride (n=370)	Haloperidol (n=118)	Total (n=488)	P
Age (mean ± SD years, range)	36.8 ± 10.9 (18–65)	39.6 ± 11.2 (20–65)	37.5 ± 11.1 (18–65)	0.015*
Males (n, %)	254 (69%)	73 (62%)	327 (67%)	0.179
Weight (mean ± SD kg, range)	72.1 ± 14.4 (42–128)	72.0 ± 15.1 (45–118)	72.1 ± 14.6 (42–128)	0.906
Racial origin (n, %)				0.970
Caucasian	357 (96%)	114 (97%)	471 (97%)	
Black	5 (1%)	1 (1%)	6 (1%)	
Asian	1 (0%)		1 (0%)	
Other	7 (2%)	3 (3%)	10 (2%)	
Type of schizophrenia (n, %)				0.205
Disorganised	122 (33%)	32 (27%)	154 (32%)	
Paranoid	160 (43%)	53 (45%)	213 (44%)	
Undifferentiated	88 (24%)	33 (28%)	121 (25%)	
Course (n, %)				0.515
Subchronic	37 (10%)	6 (5%)	43 (9%)	
Chronic	332 (90%)	111 (94%)	443 (91%)	
Duration of illness (mean ± SD years)	12 ± 8.4	13 ± 8.7	12 ± 8.5	0.026*
Patients hospitalised at baseline (n, %)	227 (75%)	95 (81%)	372 (76%)	0.263
BPRS score (mean ± SD)	55.9 ± 12.5	57.3 ± 12.6	56.2 ± 12.6	0.269
Simpson–Angus scale score (mean ± SD)	0.35 ± 0.5	0.42 ± 0.56	0.37 ± 0.52	0.276
Global akathisia absent or questionable (n, %)	330 (89%)	99 (84%)	429 (88%)	0.259
AIMS total score (mean ± SD)	2.3 ± 4.6	2.9 ± 5.2	2.5 ± 4.8	0.049*

*Significant, Kruskal–Wallis and chi-squared tests.

Table 2. Reasons for withdrawal from the study

Reason (*n*, %)	Amisulpride (*n*=370)	Haloperidol (*n*=119)
Lost to follow-up	10 (3%)	6 (5%)
Uncooperativeness	79 (21%)	17 (14%)
Lack of efficacy*	33 (9%)	20 (17%)
Adverse events	30 (8%)	12 (10%)
Recovery	1 (<1%)	0
Other	14 (4%)	7 (6%)

*Refers both to primary lack of efficacy or to relapse after improvement.

haloperidol, 15.1 ± 7.3 mg. By the study end, mean daily consumption had stabilised at 605 ± 267 mg of amisulpride and 14.6 ± 6.8 mg of haloperidol.

In the amisulpride group, 254/370 (69%) patients experienced at least one treatment-emergent adverse event, as did 82/118 (70%) patients receiving haloperidol (Table 3). The most frequently reported adverse events were psychiatric and extrapyramidal in nature. In particular, a markedly greater proportion of patients in the haloperidol group experienced an extrapyramidal disorder than in the amisulpride group (28% versus 13%). The frequency of the other most common adverse events was otherwise similar in each group with the exception of amenorrhoea and weight increase, both of which were more frequent in the amisulpride group (Table 3). The weight assessments showed a higher incidence of amisulpride-treated patients (32% versus 18%) with a weight increase above 5% of initial weight.

Serious adverse events were reported in 38/370 (10%) patients receiving amisulpride and in 8/118 (7%) patients in the haloperidol group (Table 3). These events were largely of a psychiatric nature in both groups, being related to the patients' schizophrenia (psychosis, agitation, aggression and suicide attempts). Psychiatric disorders were also the principal reason for discontinuation from the study in the amisulpride group, whereas an extrapyramidal disorder was the most frequent adverse event leading to discontinuation in the haloperidol group (Table 3).

The male:female ratio was comparable in each treatment group (M:F 2:1) and for the majority of treatment-emergent adverse events no gender effects were noted. There was a trend towards a slightly greater proportion of female patients in each group having at least one treatment-emergent adverse event, extrapyramidal disorders and, as might be expected, endocrine disorders. The effects of age were unassessable due to the small numbers of elderly patients recruited.

The clinical chemistry and haematology results were generally unremarkable and comparable in each group. Amisulpride induced a three-fold increase in prolactin levels from baseline to the end of the study versus a 1.5-fold increase in the haloperidol group although the overall incidence of endocrine adverse events was not notably different between the groups (Table 3).

Haemodynamic parameters were not significantly affected by either treatment. No remarkable effect of

Table 3. Distribution of emergent adverse events (incidence ≥ 5%) in the intention-to-treat population (all patients having received at least one dose of study treatment)

	Amisulpride (*n*=370)	Haloperidol (*n*=118)
No. of patients (*n*, %) with at least		
One treatment-emergent adverse event	254 (69%)	82 (70%)
One extrapyramidal symptom	96 (26%)	48 (41%)
One endocrine disorder	15 (4%)	3 (3%)
One serious adverse event	38 (10%)	8 (7%)
No. of patients (*n*, %) with		
Insomnia	50 (14%)	15 (13%)
Anxiety	42 (11%)	17 (14%)
Psychosis	22 (6%)	4 (3%)
Agitation	15 (4%)	6 (5%)
Extrapyramidal disorder*	48 (13%)	33 (28%)
Hyperkinesia	33 (9%)	11 (9%)
Tremor	19 (5%)	7 (6%)
Dyskinesia	9 (2%)	6 (5%)
Weight increase	40 (11%)	4 (3%)
Amenorrhoea (female patients)	7 (6%)	0
Discontinuations due to adverse events	30 (8%)	12 (10%)

*Not otherwise specified.

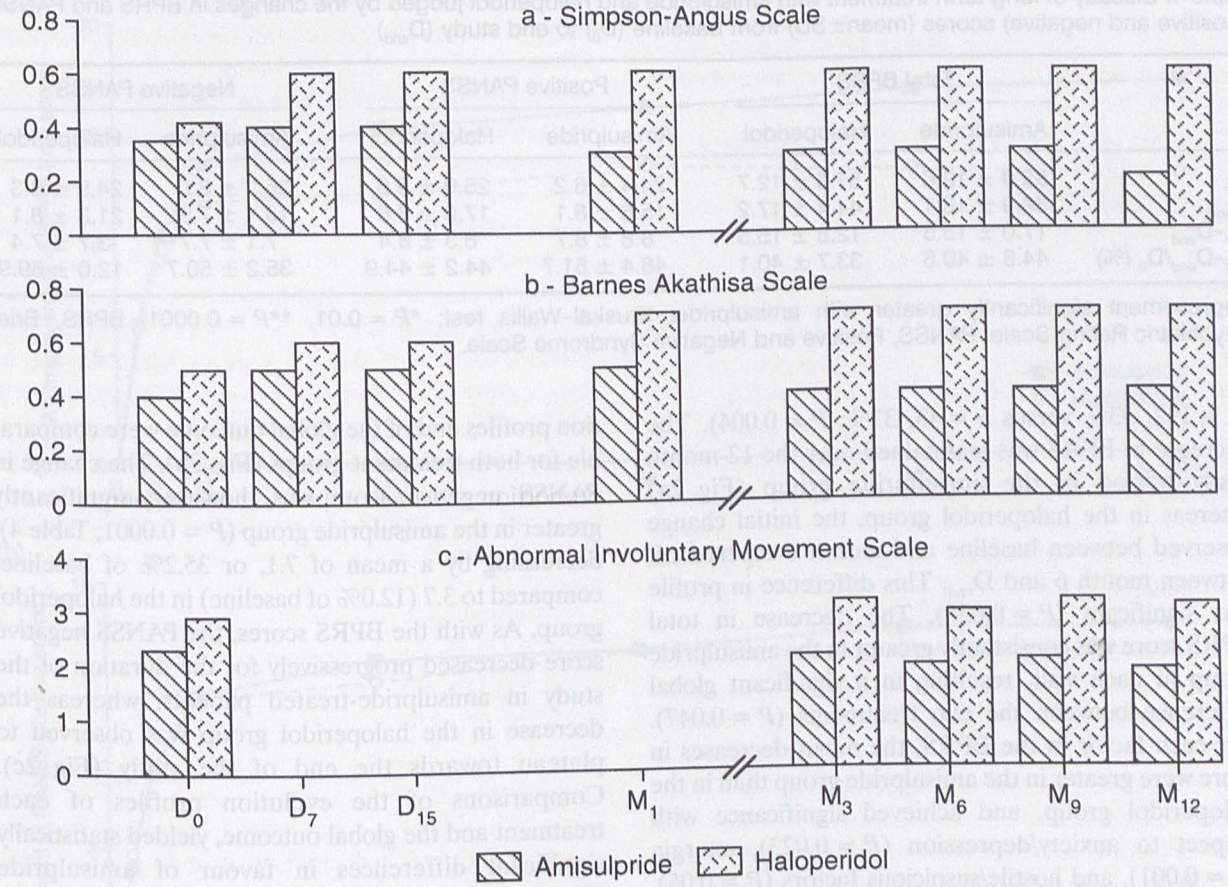


Figure 1. Mean improvement at each visit using the LOCF (last observation carried forward) procedure of the (a) Simpson–Angus Scale, (b) Barnes Akathisia Scale and (c) Abnormal Involuntary Movement Scale. Amisulpride, $n = 346$; haloperidol, $n = 111$.

either treatment was observed on the ECG results, and especially no QTc value above 500 ms was recorded and neither treatment was associated with rhythm abnormalities.

The neurological safety of amisulpride was generally superior to that of haloperidol as judged by the scores for each of the neurological rating scales throughout the study (Fig. 1). Evaluation of the Simpson–Angus scale scores showed that both the maximal aggravation and change in intensity from baseline of extrapyramidal symptoms were significantly greater in the haloperidol group ($P = 0.0001$). At the end of the study, haloperidol-treated patients had a mean change from baseline score of – 0.22, indicating aggravation, while a positive mean change of 0.11 in the amisulpride group showed reduction in extrapyramidal symptoms. As a consequence, the requirement for antiparkinsonian medication was markedly less in the amisulpride group (118/370, 32%) compared to the haloperidol group (66/118, 56%). Global clinical assessment of akathisia showed that more patients in the amisulpride group were free from signs of akathisia compared to haloperidol (amisulpride 336/370, 91% versus haloperidol 92/118, 78%; $P < 0.0001$). Changes in AIMS mean scores from baseline to study end were also significantly different between groups, reflecting an improvement in the amisulpride patients (mean change 0.6, $P = 0.014$) and a slight deterioration in the haloperidol group (mean change – 0.2).

Efficacy

At the end of the 12-month treatment period, the improvement in BPRS score was significantly greater ($P = 0.01$) in the patients receiving amisulpride than in the haloperidol group (Table 4). A mean change of 17.0 (44.8% of baseline) was computed for the amisulpride patients compared to 12.8 (33.7% of baseline) in patients receiving haloperidol, and the proportion of patients showing a 50% improvement in total BPRS score was significantly greater in the amisulpride group

Table 4. Efficacy of long-term treatment with amisulpride and haloperidol judged by the changes in BPRS and PANSS (positive and negative) scores (mean±SD) from baseline (D_0) to end study (D_{end})

	Total BPRS		Positive PANSS		Negative PANSS	
	Amisulpride	Haloperidol	Amisulpride	Haloperidol	Amisulpride	Haloperidol
D_0	55.9 ± 12.6	57.3 ± 12.7	24.4 ± 6.2	25.9 ± 6.6	25.2 ± 8.0	24.9 ± 8.3
D_{end}	38.9 ± 16.1	44.4 ± 17.2	15.6 ± 8.1	17.6 ± 8.8	18.1 ± 7.9	21.3 ± 8.1
D_0–D_{end}	17.0 ± 15.8*	12.8 ± 15.5	8.8 ± 8.7	8.3 ± 8.4	7.1 ± 7.7**	3.7 ± 7.4
D_0–D_{end}/D_0 (%)	44.8 ± 40.6	33.7 ± 40.1	48.4 ± 51.7	44.2 ± 44.9	35.2 ± 50.7	12.0 ± 69.9

Improvement significantly greater with amisulpride, Kruskal–Wallis test; *P = 0.01, **P = 0.0001. BPRS, Brief Psychiatric Rating Scale; PANSS, Positive and Negative Syndrome Scale.

(n = 192, 53% versus n = 43, 37%; P = 0.004). The decrease in BPRS was maintained over the 12-month study period in the amisulpride group (Fig. 2a) whereas in the haloperidol group, the initial change observed between baseline and month 6 tapered off between month 6 and D_{end}. This difference in profile was significant (P = 0.026). The decrease in total BPRS score was consistently greater in the amisulpride group at each visit, resulting in a significant global difference between the two treatments (P = 0.047). For each factor in the BPRS, the mean decreases in score were greater in the amisulpride group than in the haloperidol group, and achieved significance with respect to anxiety/depression (P = 0.023), anergia (P = 0.001), and hostile/suspicious factors (P = 0.05). Using a different approach, a subpopulation of patients having responded (BPRS improvement at least 20% of baseline) after 1 month of treatment (253 in the amisulpride and 69 in the haloperidol group, corresponding to 68% and 58% of the initial population) was identified. A conservative estimation of maintenance of effect, considering all discontinuations and the lack of maintenance of the 20% BPRS improvement as a treatment failure, showed similar results in both groups: 59% (149/253) of the amisulpride responders and 55% (38/69) of the haloperidol responders maintained efficacy. This estimation is comparable to the results in a long-term study comparing sertindole with haloperidol (Daniel *et al.*, 1998) using similar criteria. The data from long-term extension studies with olanzapine (Dellva *et al.*, 1997; Tran *et al.*, 1998) give lower response rates, but in these studies failure of maintenance was only defined by rehospitalisation, which is expected to be much less frequent than aggravation of symptomatology and not exclusively determined by the clinical state of the patient.

Mean decreases in the PANSS positive scores computed at the end of the study were similar in each group (P = 0.311; Table 4). Comparisons of the evolution profiles and of the global outcome were comparable for both treatment groups (Fig. 2b). The change in PANSS negative score was, however, significantly greater in the amisulpride group (P = 0.0001; Table 4), decreasing by a mean of 7.1, or 35.2% of baseline, compared to 3.7 (12.0% of baseline) in the haloperidol group. As with the BPRS scores, the PANSS negative score decreased progressively for the duration of the study in amisulpride-treated patients, whereas the decrease in the haloperidol group was observed to plateau towards the end of the study (Fig. 2c). Comparisons of the evolution profiles of each treatment and the global outcome, yielded statistically significant differences in favour of amisulpride (P < 0.001).

The number of patients rated as 'very much improved' or 'much improved' using the CGI-2 index was greater in the amisulpride than in the haloperidol group (n = 200, 55% versus n = 52, 44%), just missing significance (P = 0.057). Likewise, the improvement in overall QLS score from baseline to end of study was significantly greater among amisulpride-treated patients (−0.64) compared to the haloperidol group (−0.30) (P = 0.02). The analysis revealed a significantly greater improvement in the amisulpride group for three of the four subscores in the QLS scale – interpersonal relations (P = 0.003), instrumental role (P = 0.04) and intrapsychic foundations (P = 0.012). Statistical significance was also achieved by comparison of the change in mean GAF scores, which were −20.1 for amisulpride and −13.6 for haloperidol (P = 0.001).

The proportion of patients discontinuing from the study due to lack of efficacy (including those lost to follow-up) was significantly lower in the amisulpride group (n = 43, 12%) than in the haloperidol group (n = 26, 22%) (P = 0.009). The requirement for concomitant benzodiazepines was similar in each group (amisulpride n = 210, 57%; haloperidol n = 69, 58%).

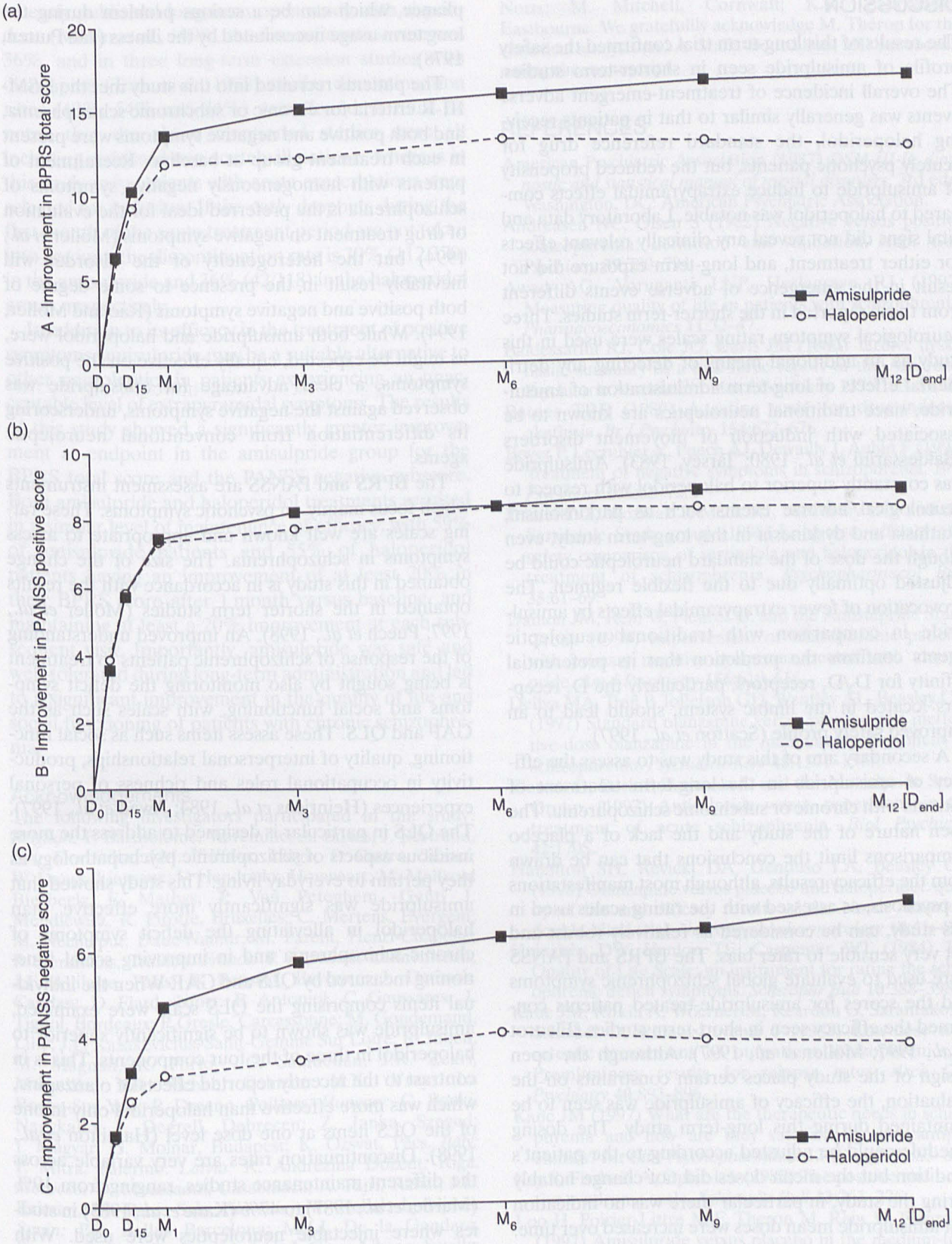


Figure 2. Mean improvement at each visit ($D_0 - D_{end}$) using the LOCF (last observation carried forward) procedure in the (a) BPRS total score, (b) PANSS positive score, and (c) PANSS negative score. Amisulpride, $n = 365$; haloperidol, $n = 117$.

DISCUSSION

The results of this long-term trial confirmed the safety profile of amisulpride seen in shorter-term studies. The overall incidence of treatment-emergent adverse events was generally similar to that in patients receiving haloperidol, the standard reference drug for acutely psychotic patients, but the reduced propensity of amisulpride to induce extrapyramidal effects compared to haloperidol was notable. Laboratory data and vital signs did not reveal any clinically relevant effects for either treatment, and long-term exposure did not result in the emergence of adverse events different from those reported in the shorter-term studies. Three neurological symptom rating scales were used in this study as an additional means of detecting any detrimental effects of long-term administration of amisulpride, since traditional neuroleptics are known to be associated with induction of movement disorders (Baldessarini *et al.*, 1980; Tarsey, 1983). Amisulpride was constantly superior to haloperidol with respect to neurological adverse events such as parkinsonism, akathisia and dyskinesia in this long-term study, even though the dose of the standard neuroleptic could be adjusted optimally due to the flexible regimen. The provocation of fewer extrapyramidal effects by amisulpride in comparison with traditional neuroleptic agents confirms the prediction that its preferential affinity for D_2/D_3 receptors, particularly the D_3 receptors located in the limbic system, should lead to an improved safety profile (Scatton *et al.*, 1997).

A secondary aim of this study was to assess the efficacy of amisulpride in the long-term treatment of patients with chronic or subchronic schizophrenia. The open nature of the study and the lack of a placebo comparisons limit the conclusions that can be drawn from the efficacy results, although most manifestations of psychosis, as assessed with the rating scales used in this study, can be considered as relatively robust and not very sensible to rater bias. The BPRS and PANSS were used to evaluate global schizophrenic symptoms and the scores for amisulpride-treated patients confirmed the efficacy seen in short-term studies (Fleurot *et al.*, 1997; Möller *et al.*, 1997). Although the open design of the study places certain constraints on the evaluation, the efficacy of amisulpride was seen to be maintained during this long-term study. The dosing schedule could be adjusted according to the patient's condition but the mean doses did not change notably during the study, in particular there was no indication that amisulpride mean doses were increased over time. This flexible dosage approach to the management of schizophrenia, based on the stage of the disease, has significant potential for improvement of patient compliance, which can be a serious problem during the long term usage necessitated by the illness (van Putten, 1978).

The patients recruited into this study met the DSM-III-R criteria for chronic or subchronic schizophrenia, and both positive and negative symptoms were present in each treatment group at baseline. Recruitment of patients with homogeneously negative symptoms of schizophrenia is the preferred ideal for the evaluation of drug treatment on negative symptoms (Möller *et al.*, 1994) but the heterogeneity of the disorder will inevitably result in the presence to some degree of both positive and negative symptoms (Rao and Möller, 1994). While both amisulpride and haloperidol were, as might be expected, equally effective on the positive symptoms, a clear advantage for amisulpride was observed against the negative symptoms, underscoring its differentiation from conventional neuroleptic agents.

The BPRS and PANSS are assessment instruments which focus mainly on psychotic symptoms. These rating scales are well known and appropriate to assess symptoms in schizophrenia. The size of the change obtained in this study is in accordance with the results obtained in the shorter term studies (Möller *et al.*, 1997; Puech *et al.*, 1998). An improved understanding of the response of schizophrenic patients to treatment is being sought by also monitoring the deficit symptoms and social functioning, with scales such as the GAF and QLS. These assess items such as social functioning, quality of interpersonal relationships, productivity in occupational roles and richness of personal experiences (Heinrichs *et al.*, 1984; Awad *et al.*, 1997). The QLS in particular is designed to address the more insidious aspects of schizophrenic psychopathology as they pertain to everyday living. This study showed that amisulpride was significantly more effective than haloperidol in alleviating the deficit symptoms of chronic schizophrenia and in improving social functioning measured by QLS and GAF. When the individual items comprising the QLS scale were examined, amisulpride was shown to be significantly superior to haloperidol in three of the four components. This is in contrast to the recently reported effects of olanzapine, which was more effective than haloperidol only in one of the QLS items at one dose level (Hamilton *et al.*, 1998). Discontinuation rates are very variable across the different maintenance studies, ranging from 19% (Marder *et al.*, 1987) to 47% (Kane *et al.*, 1983) in studies where injectable neuroleptics were used. With respect to these data, the discontinuation rate of 45% in the amisulpride group in a study where oral dosing was used, does not seem to be notably different. In a

recently published long-term open study with risperidone (Möller *et al.*, 1998), the discontinuation rate was 36%, and in three long-term extension studies with olanzapine (Tran *et al.*, 1998) similar discontinuation rates (40%, 54% and 34%) were found. It must be noted that the above mentioned studies were all including stabilised non acutely ill patients, whereas in this study only patients with acute exacerbations were selected for inclusion. If the early dropouts during the first month of the acute treatment period are not taken into account, the discontinuation rate is 30% (112/370) in the amisulpride and 36% (42/118) in the haloperidol group, respectively.

In addition to its efficacy in the treatment of positive symptoms, amisulpride may be a suitable alternative to other neuroleptics in patients experiencing an unacceptable level of extrapyramidal symptoms. The results of this study showed a significantly greater improvement at endpoint in the amisulpride group for the BPRS total score and the PANSS negative subscore. Both amisulpride and haloperidol treatments resulted in a similar level of maintenance of efficacy, with 59% of amisulpride patients and 55% of haloperidol patients having an improvement of at least 20% in their BPRS score after 1 month versus baseline, and maintaining at least a 20% improvement at each subsequent visit. Importantly, amisulpride was safe and well-tolerated during long-term administration and led to a significant improvement in the quality of life and social functioning of patients with chronic schizophrenia.

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